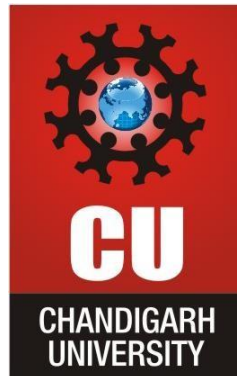




CHANDIGARH UNIVERSITY

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Subject: 24CSH-337

Subject code: Computer Networks

Assignment No.: 2

Student Name: Husanpreet Kaur

UID: 24BCY70138

Branch: AIT-CSE (Cyber Security)

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Q1: What is an IP Address? Why do we need classes in case of IPv4 address? Explain each class. Mention how many hosts and networks are possible in each class. Also explain subnetting and supernetting with examples.

Sol: (i) An IP Address (Internet Protocol Address) is a logical address assigned to a network interface of a device connected to an IP network. It is used to identify the source and destination of packets and enables communication between devices across interconnected networks.

The Network Layer, which is Layer 3 of the OSI model, is responsible for logical addressing and routing. Routers use destination IP addresses to determine where packets should be forwarded.

Eg: 192.168.1.10

The MAC address of its network interface may remain the same, while its IP address changes according to the network to which it is connected.

(ii) IPv4 (Internet Protocol Version 4) uses a 32-bit address.

These 32 bits are divided into four groups of 8 bits called octets.

Eg: 192.168.1.10

In Binary: 192 168 1 10

11000000.10101000.00000001.00001010

Therefore, IPv4 Address = 32 bits

$$= 4 \times 8 \text{ bits} = 4 \text{ octets}$$

Each octet can contain a value between: 0 to 255

Because 8 bits can represent: 00000000=0 and 11111111=255

(iii) An IPv4 address can conceptually be divided into:

1. Network part
2. Host part

The network part identifies the network to which the device belongs.

The host part identifies a particular host/interface within that network.

Eg: 192.168.1.10/24 means first 24 bits identify the network.

Therefore, Network bits=24

and Host bits= 32-24=8

The corresponding subnet mask=255.255.255.0

Here, Network Address = 192.168.1.0

Broadcast Address = 192.168.1.255

Usable Host Range = 192.168.1.1 – 192.168.1.254

(iv) Why are classes needed in IPv4?

Originally, IPv4 used classful addressing.

Different organizations require different numbers of IP addresses.

If every organization received the same-sized network, a large number of IP addresses would be wasted.

Therefore, IPv4 addresses were divided into different classes according to the size of the network.

The original five classes were:

Class	Purpose
Class A	Very large networks
Class B	Medium-sized networks
Class C	Small networks
Class D	Multicast
Class E	Experimental/Reserved

(v) Class A- It was designed for very large networks.

Address Range:

1.0.0.0 to 126.255.255.255

The first octet ranges from:

1 to 126

The default subnet mask is:

255.0.0.0 or /8

Class A address has:

1 bit → Class identification

7 bits → Network ID

24 bits → Host ID

Therefore:

0 | Network ID | Host ID

There are 24 host bits.

Number of Networks-

There are 7 variable network bits:

$$2^7 = 128$$

However:

- Network 0 is reserved.
- Network 127 is reserved for loopback.

Therefore:

$$128 - 2 = 126 \text{ networks}$$

Number of Hosts-

There are 24 host bits.

Therefore:

$$2^{24} = 16,777,216 \text{ addresses}$$

Two addresses cannot normally be assigned to hosts:

- All host bits = 0 → Network address
- All host bits = 1 → Broadcast address

Therefore:

$$2^{24} - 2$$

$$= 16,777,214 \text{ usable hosts}$$

Thus:

Class A provides 126 networks with 16,777,214 usable hosts per network.

Why is 127 excluded?

The address range:

127.0.0.0/8

is reserved for loopback.

For example:

127.0.0.1

refers to the local host.

(vi) Class B- It was designed for medium-sized networks.

Address Range: 128.0.0.0 to 191.255.255.255

First octet: 128 – 191

Default subnet mask: 255.255.0.0 or /16

Class B starts with binary: 10

Therefore:

2 bits → Class identification

14 bits → Network ID

16 bits → Host ID

Thus there are:

16 network bits

16 host bits

Number of Networks-

There are 14 variable network bits:

$2^{14} = 16,384$ networks

Number of Hosts-

There are 16 host bits:

$2^{16} = 65,536$ addresses

Two addresses are reserved:

$65,536 - 2 = 65,534$ usable hosts

Therefore:

Class B provides 16,384 networks with 65,534 usable hosts per network.

(vii) Class C- It was designed for smaller networks.

Address Range: 192.0.0.0 to 223.255.255.255

First octet: 192 – 223

Default subnet mask: 255.255.255.0 or /24

Class C starts with: 110

3 bits → Class identification

21 bits → Network ID

8 bits → Host ID

Thus, Network bits = 24

Host bits = 8

Number of Networks-

There are 21 variable network bits:

$2^{21} = 2,097,152$ networks

Number of Hosts-

There are 8 host bits:

$2^8 = 256$ addresses

Two are reserved:

$256 - 2 = 254$ usable hosts

Therefore:

Class C provides 2,097,152 networks with 254 usable hosts per network.

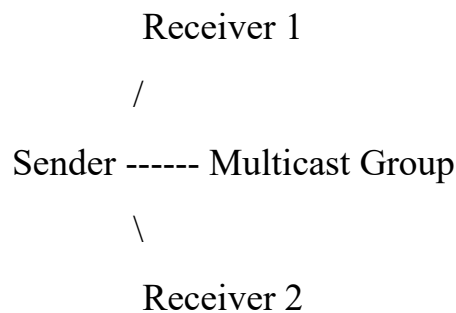
(viii) Class D- It is used for multicast communication.

Range: 224.0.0.0 to 239.255.255.255

Class D does not follow the normal network-ID/host-ID structure used by Classes A, B and C.

Multicast- It allows one sender to transmit data to multiple receivers belonging to a multicast group.

For example:



Therefore, the conventional calculation of networks and usable hosts used for Classes A, B and C is not applicable to Class D.

(ix) Class E- Its range lies from:

240.0.0.0 to 255.255.255.255

They are reserved for experimental and special purposes and are not used as ordinary host addresses.

Class	Range	Default Mask	Network Bits	Host Bits	Networks	Usable Hosts
A	1–126	/8	7 variable	24	126	16,777,214
B	128–191	/16	14 variable	16	16,384	65,534
C	192–223	/24	21 variable	8	2,097,152	254
D	224–239	—	—	—	Multicast	—
E	240–255	—	—	—	Experimental	—

(x) Subnetting: It is the process of dividing one large IP network into multiple smaller logical networks called subnets.

The main purpose of subnetting is to use IP addresses efficiently and organize a network into smaller sections.

For example, an organization may have separate departments:



(xi) Why is Subnetting required?

1. Efficient IP Address Utilization

Instead of assigning a very large network to a small department, the network can be divided according to actual requirements.

2. Reduced Broadcast Traffic

A broadcast in one subnet does not need to reach every device in all other subnets.

3. Better Network Management

Different departments can have separate subnets.

4. Improved Security

Network segmentation can help isolate different groups of devices and support security policies.

5. Easier Troubleshooting

Smaller networks are easier to monitor and troubleshoot.

(xii) How Does Subnetting Work?

Subnetting works by borrowing bits from the host portion and using them as additional subnet/network bits.

Consider:

192.168.10.0/24

Initially:

Network bits = 24

Host bits = 8

Number of addresses:

$$2^8 = 256$$

Usable hosts:

$$256 - 2 = 254$$

Suppose we borrow 2 host bits.

New prefix:

$$/24 + 2 = /26$$

Now:

Network bits = 26

Host bits = 6

Number of Subnets

$$2^2 = 4 \text{ subnets}$$

Hosts per Subnet

$$2^6 = 64 \text{ addresses}$$

Usable hosts:

$$64 - 2 = 62 \text{ hosts}$$

A /24 network divided using 2 borrowed bits produces 4 /26 subnets, each having 62 usable hosts.

Eg: 192.168.10.0/24 - network

New prefix - /26

Subnet mask – 255.255.255.192

Block size – 256-192=64

Subnet	Network Address	Usable Host Range	Broadcast
1	192.168.10.0	192.168.10.1 – 62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 126	192.168.10.127

3	192.168.10.128	192.168.10.129 – 190	192.168.10.191
4	192.168.10.192	192.168.10.193 – 254	192.168.10.255

(xiii) Supernetting: It is the process of combining multiple smaller contiguous networks into a larger logical network.

It is particularly useful for route aggregation, where several routes can be represented by a single summarized route.

Eg: Consider four networks:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Each /24 network contains:

256 addresses

Four networks contain:

$4 \times 256 = 1024$ addresses

These four contiguous networks can be summarized as:

192.168.0.0/22

A /22 network has:

$32 - 22 = 10$ host bits

Therefore:

$2^{10} = 1024$ addresses

So:

$4 \times /24$ networks

↓

one /22 network

Advantages of Supernetting:

1. Reduces routing table size.

2. Reduces routing information.
 3. Reduces routing overhead.
 4. Makes routing more efficient.
 5. Provides route summarization.
 6. Helps improve scalability of large networks.
-

Q2: What is the difference between Link State and Distance Vector Routing? Explain how these algorithms help in identifying the least-cost path with examples.

Sol: (i) Routing: It is the process of determining a suitable path through a network from a source to a destination.

A router may have several possible paths to reach a destination. The router must determine which path has the lower cost.

Routing protocols use routing metrics to compare paths.

Common metrics include:

- Hop count
- Bandwidth
- Delay
- Reliability
- Load
- Administrative cost

(ii) Least-Cost Path: A least-cost path is a path from source to destination whose total routing cost is minimum.

Suppose:

A ----2---- B

| |

5 3

| |

C ----2---D

We want to travel from A to D.

Path 1: A-> B-> D

Cost: $2+3=5$

Path 2: A-> C-> D

Cost: $5+2=7$

Thus, least-cost path: A->B->D

And total cost: 5

(iii) Distance Vector Routing: It is a routing approach in which each router maintains information about the distance/cost to destinations and the direction or next hop through which those destinations can be reached.

Routers exchange routing information with their neighboring routers.

Distance Vector uses the Bellman-Ford algorithm.

(iv) Bellman-Ford Algorithm: The Bellman-Ford approach calculates the cost of reaching a destination through each neighboring router.

Basic Equation: $D(x,y) = \min [C(x,v) + D(v,y)]$

where:

- x = current router
- y = destination
- v = neighboring router
- $C(x,v)$ = cost from x to neighbor v
- $D(v,y)$ = neighbor's cost to destination y

The router chooses the minimum result.

Eg: $A \rightarrow B = 12$

$A \rightarrow C = 9$

$C \rightarrow B = 1$

Initially: $A \rightarrow B = 12$

After A receives information from C:

$A \rightarrow C \rightarrow B$

$= A \rightarrow C + C \rightarrow B$

$= 9 + 1$

$= 10$

A compares: Direct path=12

Via C=10

Therefore, $\min(12,10)=10$

Thus, Routing table contains: destination-B

Cost-10

Next hop-C

(v) Characteristics of Distance Vector:

Advantages-

1. Relatively simple to implement.
2. Requires less complete topology information.
3. Routers exchange information with neighbors.
4. Suitable for smaller/simple networks.

Disadvantages-

1. Convergence can be slower.
2. Routing loops can occur.
3. It can suffer from the count-to-infinity problem.
4. Routing information may initially be incomplete.

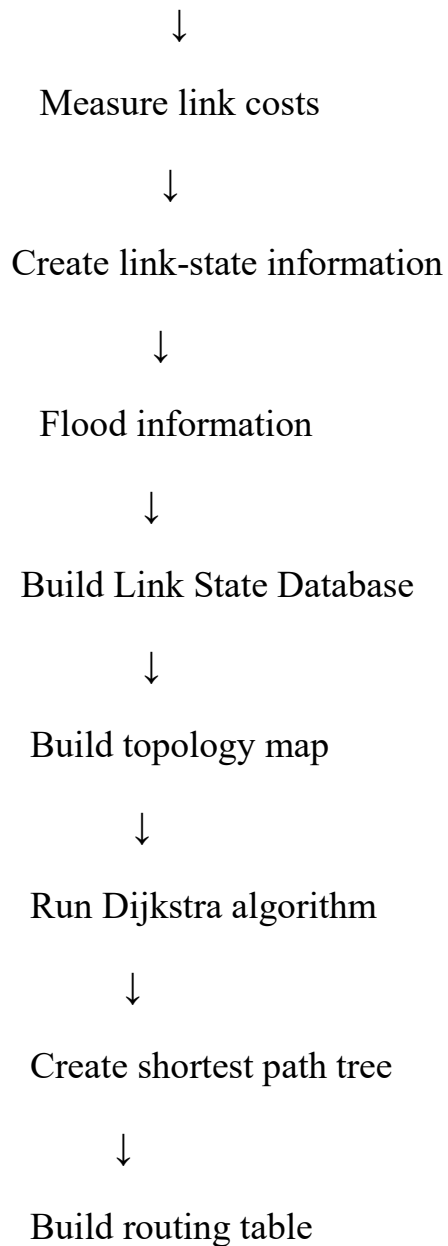
(vi) Link State Routing: In Link State Routing, each router learns information about the links and their costs throughout the network.

Instead of only learning:

"My neighbor says destination X is 5 units away."

the router builds information about the network topology.

Process: Discover neighbors



(vii) Link State Database: Each router maintains a Link State Database (LSDB) containing information about the network's topology.

The router uses this information to construct a network graph.

Eg: A-B = 4

A-C = 5

B-D = 3

$$C-D = 2$$

4

A ----- B

| |

|5 |3

| |

C ----- D

2

(viii) Dijkstra Algorithm: Link State routing generally uses Dijkstra's Shortest Path First algorithm.

The algorithm works by maintaining the currently known minimum cost from the source to each node.

Initially, source=0

Other nodes = ∞

Then:

1. Select the unvisited node with the smallest cost.
2. Examine its neighboring nodes.
3. Calculate the cost through the selected node.
4. Update the neighboring node if a lower cost is found.
5. Mark the selected node as visited.
6. Continue until the required paths are determined.

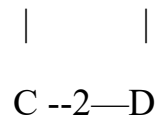
The update rule is:

$$D(v) = \min[D(v), D(u) + C(u,v)]$$

Eg for Link State: A --4--B and C—1—E, D—1—E

| |

5 3



Path 1: A->B->D->E

Cost: $4+3+1=8$

Path 2: A->C->E

Cost: $5+1=6$ (least cost)

Basis	Distance Vector	Link State
Basic principle	Shares distance information with neighbors	Shares link-state/topology information
Algorithm	Bellman-Ford	Dijkstra
Network knowledge	Mainly neighbor-based information	Maintains topology information
Information exchange	With neighboring routers	Link-state information is flooded
Calculation	Calculates routes through neighbors	Calculates shortest paths over topology
Convergence	Generally slower	Generally faster
Routing loops	More susceptible	Less susceptible
Complexity	Simpler	More complex
Example protocol	RIP	OSPF
Main idea	"How far is the destination?"	"What does the network topology look like?"

Q3: What is congestion in a network? Why do we need to control congestion? Explain Leaky Bucket and Token Bucket with examples. Also explain why fragmentation is needed.

Sol: (i) Network congestion occurs when the amount of traffic entering a network or router is greater than the network's ability to process and forward that traffic.

Congestion occurs when too many packets compete for limited network resources.

Eg: Incoming traffic = 100 Mbps

Outgoing link capacity = 10 Mbps

The router cannot immediately forward all incoming traffic.

Therefore, packets begin accumulating in the router's buffer.

Eg: Incoming=100 packets/second

Outgoing=60 packets/second

Difference: $100 - 60 = 40$ packets/second

Therefore, 40 packets per second accumulate in the buffer.

After several seconds, the buffer may become full.

When the buffer becomes full: New packets → Buffer full → Packets dropped

(ii) Causes of Congestion:

1. Excessive Traffic

Too many devices transmit data simultaneously.

2. Limited Bandwidth

The link cannot carry all incoming traffic.

3. Limited Router Processing Capacity

A router may not process packets fast enough.

4. Limited Buffer Space

The router has limited memory for queued packets.

5. Bursty Traffic

A sudden burst of packets may temporarily exceed network capacity.

6. Network Failures

When a link fails, traffic may be redirected through another link, increasing its load.

(iii) Effects of Congestion: (a) Increased Delay: Packets wait longer in queues.

(b) Packet Loss: Packets are discarded when buffers become full.

(c) Retransmissions: Lost packets may be retransmitted.

(d) Reduced Throughput: The amount of useful data successfully delivered can decrease.

(e) Buffer Overflow: The available queue space becomes exhausted.

(f) Poor Application Performance: Applications such as video conferencing, online gaming and voice communication may experience poor performance.

(iv) Why is Congestion Control Needed?

Congestion control is required to prevent or reduce excessive traffic so that the network can operate efficiently.

Without congestion control:

Excess traffic



Queue increases



Buffer fills



Packet loss



Retransmissions



More traffic



More congestion

Therefore, congestion control helps to:

1. Reduce packet loss.
2. Reduce network delay.
3. Prevent buffer overflow.
4. Improve throughput.
5. Maintain network stability.
6. Improve application performance.
7. Control bursty traffic.

(v) Traffic Shaping: It is a technique used to control the rate and pattern at which packets enter the network.

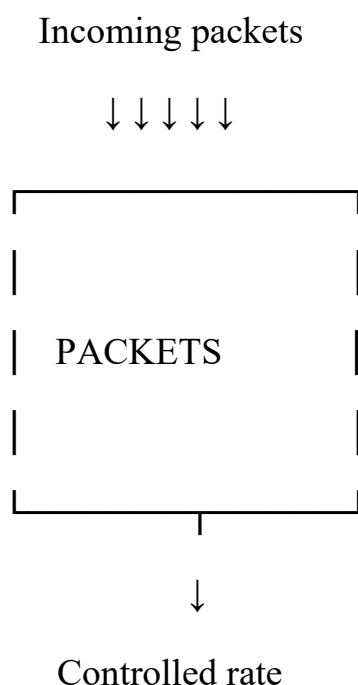
The objective is to prevent sudden bursts of traffic from overwhelming network resources.

Two important traffic-shaping techniques are:

1. Leaky Bucket
2. Token Bucket

(vi) Leaky Bucket Algorithm: It controls traffic by storing incoming packets in a finite buffer and transmitting them at a controlled rate.

It can be imagined as a bucket with a small hole at the bottom.



Working: 1. Packets arrive at the bucket.

2. Packets are stored in a buffer.

3. Packets leave the bucket at a fixed/controlled rate.

4. If the bucket becomes full, newly arriving packets may be discarded.

Suppose: Bucket capacity = 10 packets

Packets currently stored = 7

Available space: $10 - 7 = 3$ packets

If 5 new packets arrive:

Capacity available = 3

Packets arriving = 5

Therefore:

Accepted = 3 packets

Dropped = 2 packets

Advantages of Leaky Bucket-

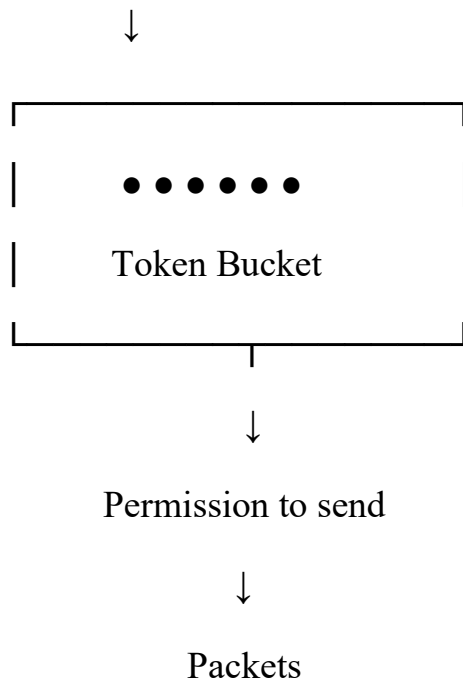
1. Smooths bursty traffic.
2. Produces a controlled output rate.
3. Prevents sudden bursts from overwhelming downstream networks.
4. Simple concept and implementation.
5. Provides predictable traffic flow.

Disadvantages of Leaky Bucket-

1. Packets may experience waiting delay.
2. Packets can be dropped when the bucket is full.
3. It does not preserve unused transmission capacity.
4. It does not naturally allow large bursts.

(vii) Token Bucket Algorithm: It controls traffic by generating tokens at a specified rate. Tokens are stored in a bucket. A packet can be transmitted when sufficient tokens are available.

Tokens generated



In a simple packet-based model: 1 token = permission to send 1 packet

Working: Token generation rate = 5 tokens/second

Bucket capacity = 10 tokens

If there is no traffic for some time, tokens accumulate.

Suppose the bucket reaches: 10 tokens

Now 8 packets arrive.

Because 8 tokens are available: 8 packets can be transmitted.

After transmission: Remaining tokens = $10 - 8 = 2$ tokens

Therefore, Token Bucket can support a controlled burst when tokens have accumulated.

Advantages of Token Bucket-

1. Allows controlled bursts.
2. Efficiently uses available capacity.
3. Tokens can accumulate during idle periods.
4. Provides flexible traffic control.
5. Useful when occasional bursts are acceptable.

Disadvantages of Token Bucket-

1. More complex than a basic Leaky Bucket model.
2. If token generation is too slow, packets may need to wait.
3. If the token bucket is full, excess tokens are discarded.

Feature	Leaky Bucket	Token Bucket
Main concept	Packets are stored and released at a controlled rate	Tokens are generated and stored
What is stored?	Packets	Tokens
Output	Generally smooth	Can be bursty when tokens have accumulated
Idle period	Unused capacity is not stored for later bursts	Tokens accumulate
Burst traffic	Strongly smooths/restricts bursts	Allows controlled bursts
Overflow	Incoming packets may be discarded	Excess tokens may be discarded
Main parameter	Output/leak rate	Token generation rate
Main advantage	Smooth output	Flexible burst handling

(viii) Fragmentation: It is the process of dividing a large IP packet into smaller pieces called fragments so that the packet can travel through a network whose MTU is smaller than the packet size.

Different networks can support different maximum packet sizes.

The maximum packet size that a network/link can carry is called the MTU (Maximum Transmission Unit).

Why is Fragmentation needed?

Suppose a packet is:

Packet size = 1200 bytes

and the next network has:

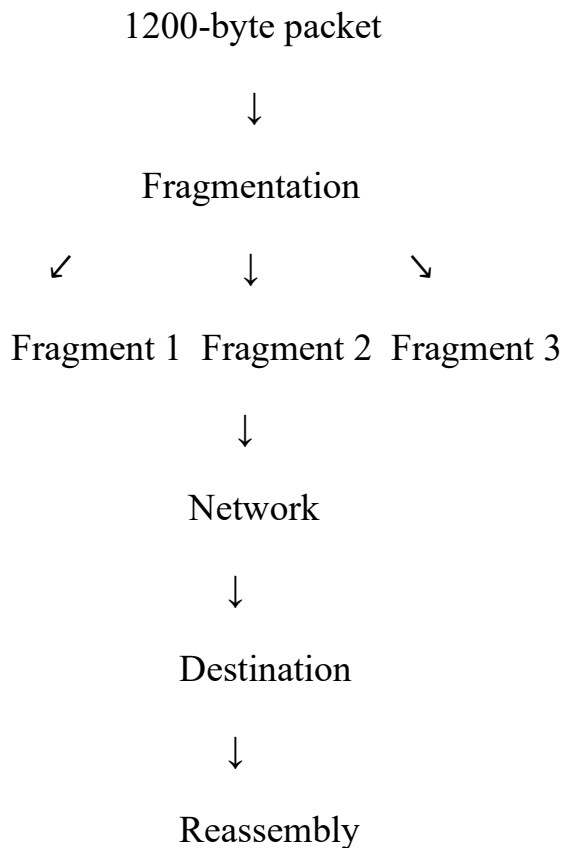
MTU = 500 bytes

The complete packet cannot fit into the network because:

$1200 > 500$

Therefore, under IPv4 fragmentation rules, the packet may need to be divided into smaller fragments.

Conceptually:



(ix) IPv4 Fragmentation-

Consider: Original IP packet = 4000 bytes

MTU = 1500 bytes

A 4000-byte packet cannot be transmitted as a single packet over a 1500-byte MTU link.

The fragments must contain sufficient information for the destination to determine which original packet they belong to and where each fragment belongs.

Important Fields Used in IPv4 Fragmentation-

Three important IPv4 header fields are involved in fragmentation:

1. Identification

The Identification field helps the destination determine which fragments belong to the same original IP datagram.

2. Flags

The Flags field contains information related to fragmentation, including whether more fragments follow.

3. Fragment Offset

The Fragment Offset identifies the position of a fragment's data relative to the original IP datagram.

For example, if a packet is divided into:

Fragment 1

Fragment 2

Fragment 3

the destination uses the fragment information to place the data in the correct order.

(x) Reassembly- After the fragments reach the destination, they are reassembled to reconstruct the original IP datagram.

The basic process is:

Original packet



Fragmentation



Fragment 1 —┐

Fragment 2 —┤—→ Destination → Reassembly

Fragment 3 —┐



Original packet reconstructed

An important point is that fragmentation and reassembly are related but occur at different locations: fragmentation may occur during IPv4 forwarding, while reassembly occurs at the destination.

(xi) IPv4 vs IPv6 Fragmentation-

IPv4: Under IPv4 rules, routers may fragment packets in transit when required by the outgoing link's MTU.

IPv6: In IPv6, routers do not fragment packets in transit.

If fragmentation is necessary, the source uses the IPv6 fragmentation mechanism.

